

§ 5.3 Riemann-Roch Theorem

- For a line bundle L

$$\text{ch}(L) = \exp(c_1(L)) = 1 + c_1(L) + \dots + \frac{1}{n!} c_1(L)^n$$

$$\text{td}(L) = \frac{c_1(L)}{1 - \exp(-c_1(L))} = 1 + \frac{1}{2} c_1(L) + \dots$$

- Every variety X has a "homology Todd class"

$$\text{Td}(X) \in A_*(X)_{\mathbb{Q}}$$

$$\text{Td}(X) = \text{Td}_n(X) + \dots + \text{Td}_0(X), \quad \text{Td}_k(X) \in A_k(X)_{\mathbb{Q}}$$

$$\text{Td}_n(X) = |X|$$

When X is nonsingular

$$\text{Td}(X) = \text{td}(T_X) \cap |X|$$

- Riemann-Roch

$\forall$ vector bundle E , variety X

$$\chi(X, E) = \int \text{ch}(E) \cap \text{Td}(X) = \deg(\text{ch}(E) \cap \text{Td}(X))_0$$

When X is nonsingular

$$\chi(X, E) = \deg(\text{ch}(E) \cap \text{Td}(T_X))_0$$

For a line bundle

$$\chi(X, L) = \sum_{k=0}^n \frac{1}{k!} \deg(c_1(L)^k \cap \text{Td}_k(X))$$

If $L = \mathcal{O}_X$

$$\chi(X, \mathcal{O}_X) = \deg(\text{Td}_0(X))$$

- If $f: X' \rightarrow X$ is proper birational
 $f_*(\mathcal{O}_{X'}) = \mathcal{O}_X$

In toric variety (non-singular)

$$\begin{aligned} \bullet \quad & \begin{cases} 0 \rightarrow \Omega_X' \rightarrow \Omega_X'(\log D) \rightarrow \bigoplus_{i=1}^d \mathcal{O}_{D_i} \rightarrow 0 \\ \Omega_X'(\log D) \text{ is trivial. } (\Omega_X'(\log D) \cong M \otimes_{\mathbb{Z}} \mathcal{O}_X) \end{cases} \\ & \Rightarrow c(\Omega_X') \cdot \prod_{i=1}^d c(\mathcal{O}_{D_i}) = 1 \quad (c = 1 + c_1 + \dots + c_n) \end{aligned}$$

$$\bullet \quad \text{From } 0 \rightarrow \mathcal{O}(-D_i) \rightarrow \mathcal{O}_X \rightarrow \mathcal{O}_{D_i} \rightarrow 0$$

$$\Rightarrow c(\mathcal{O}_{D_i}) = (1 - D_i)^{-1}$$

$$\Rightarrow c(\Omega_X') = \prod_{i=1}^d (1 - D_i)$$

$$\bullet \quad c_i(E^\vee) = (-1)^i c_i(E) \Rightarrow c_i(T_X) = (-1)^i c_i(\Omega_X')$$

$$\Rightarrow c_i(T_X) = (-1)^i (-1)^i \sum_{r_1, \dots, r_i} D_{r_1} \cdots D_{r_i}$$

$$\Rightarrow c(T_X) = \prod_{i=1}^d (1 + D_i)$$

$$= \sum_{b \in \Delta} [V(b)]$$

$$\text{td}(T_X) = \prod_{i=1}^d \frac{D_i}{1 - \exp(-D_i)}$$

$$= 1 + \frac{1}{2} c_1 + \frac{1}{24} (c_1^2 + c_2) + \frac{1}{24} c_1 c_2 + \dots$$

If $X(\Delta)$ is ~~non~~ singular,

Let Δ' be a refinement of Δ and $X(\Delta')$ is nonsing.

$f: X(\Delta') \rightarrow X(\Delta)$ is a resolution of singularities and $f_*(\mathcal{O}_{X'}) = \mathcal{O}_X$

$$\text{Rif } f_*(\mathcal{O}_{X'}) = 0 \quad \forall i > 0$$

$$\Rightarrow Td(X(\Delta)) = f_*(Td(X(\Delta')))$$

Ex: For any toric variety, $Td_{n-1}(X) = \frac{1}{2} \sum [D_i]$

$$\boxed{Td_0(X) = [X]}, \text{ for any point } x \in X.$$

$$\text{In fact: } Td_{n-1}(X) = f_*(Td_{n-1}(X(\Delta'))) = f_*(\frac{1}{2} \sum [D_i']) = \frac{1}{2} \sum [D_i]$$

General toric variety

Let $\mathcal{L} = \mathcal{O}(D)$ g.g., D T-Cartier

$$\Rightarrow h^0(X, \mathcal{L}) = \#P \quad H^i(X, \mathcal{L}) = 0 \quad i > 0$$

H-Riemann-Roch

$$\Rightarrow \chi(X, \mathcal{L}) = \#P$$

$$= \sum_{k=0}^n \frac{1}{k!} \text{degree}(D^k \wedge Td_k(X))$$

$f: \Delta' \rightarrow \Delta'$
resolution of sing

$$\chi(X, \mathcal{L}) = \#P$$

$$= \sum_{k=0}^n \frac{1}{k!} \text{degree}(D^k \wedge f^*(Td(\Delta')))$$

Is independent with the choice
of $f: \Delta' \rightarrow \Delta$?

Then we have

$$f_P(v) := \chi(X, \mathcal{O}(v \cdot D)) = \#v \cdot P$$

$$= \sum_{k=0}^n \frac{1}{k!} \text{degree}(v^k D^k \wedge Td_k(X))$$

$$= \sum_{k=0}^n a_k v^k$$

$$(a_k = \frac{1}{k!} \deg(D^k \wedge Td_k(X)))$$

(Recall: If D is ample, $a_n = \frac{D^n}{n!} \neq 0$)

$$\text{prop 1: } \text{vol}(P) = \lim_{v \rightarrow \infty} \frac{\#(v \cdot P)}{v^n} = \frac{D^n}{n!}$$

$$\text{pf: } \textcircled{1} \#v \cdot P = \sum_{k=0}^n a_k v^k \Rightarrow \lim_{v \rightarrow \infty} \frac{\#(v \cdot P)}{v^n} = \frac{D^n}{n!} \textcircled{2}$$

$$\textcircled{2} \text{vol}(P) = \lim_{v \rightarrow \infty} \frac{\#(v \cdot P)}{v^n} ?$$

Let $P_6 = P \cap (6^\perp + u(6))$

(If $\text{codim}(6) = k$, $\dim(V(6)) = k$, $\dim(P_6) = k$)

prop 2: $\text{Vol}(P_6) = \deg\left(\frac{D^k}{k!} \cap V(6)\right)$

Rmk: prop 1 is the special case of prop 2 when $6 = \{0\}$

pf: ① If $V(6) \not\subset \text{supp}(D)$

(i.e. $\langle u(6), v_i \rangle = a_i = 0 \quad \forall v_i \in 6$)

i.e. $u(6) = 0$ in $M/6^\perp$)

then $D|_{V(6)}$ is defined by the collection of $u(z)$ (or $\chi^{-u(z)}$), $u(z) \in M(6)/M(z)$ as z varies over $\text{star}(6)$

So $D|_{V(6)}$ corresponds to a polytope $6^\perp \cap P \cap M \stackrel{\downarrow}{=} P_6 \cap M(6)$

Since $u(6) = 0$, $P_6 = P \cap 6^\perp$

By prop 1, $\text{Vol}(P_6) = \deg\left(\frac{(D|_{V(6)})^k}{k!}\right)$
 $= \deg\left(\frac{D^k}{k!} \cap V(6)\right)$

② In general, D can be replaced by $D + \text{div}(X^{u(6)})$.

The polytope corresponds to $D + \text{div}(X^{u(6)})$ is $P - u(6)$

$\mathcal{O}(D|_{V(6)}) \cong \mathcal{O}(D + \text{div}(X^{u(6)}))|_{V(6)}$

and $D + \text{div}(X^{u(6)})$ corresponds to the polytope $6^\perp \cap M \cap (P - u(6))$

12

Since $\left\{ \begin{array}{l} \chi(X, \mathbb{Z}) = \#P = \sum_{k=0}^n \frac{1}{k!} \deg(D^k \cap Td_k(X)) \\ Vol(P_6) = \deg\left(\frac{D^k}{k!} \cap [V_6]\right) \\ \text{Set } Td(X) = \sum_{6 \in \Delta} \gamma_6 [V_6], \gamma_6 \in \mathbb{Q} \end{array} \right.$

(Since V_6 span $A_*(X) \otimes \mathbb{Q}$)
 (Rmk: γ_6 is not unique).

$\Rightarrow$ We have $\#(P) = \sum_{6 \in \Delta} \gamma_6 \cdot Vol(P_6)$

$\chi(X, \mathbb{Q}(V_D)) = \#V \cdot P$ and ~~$a_k = \frac{1}{k!} \deg(D^k \cap Td_k)$~~
 $= \sum_{k=0}^n a_k V^k$

and $a_k = \frac{1}{k!} \deg(D^k \cap Td_k(X))$

$\Rightarrow a_k = \sum_{\text{codim}(6)=k} \gamma_6 \cdot Vol(P_6)$

In particular, 2-dim

$Td(X) = [X] + \frac{1}{2}(\sum [D_i]) + [X]$

prop: (Pick's formula) $\#(P) = \text{Area}(P) + \frac{1}{2} \text{Perimeter}(P) + 1$

• Cup product

a product on the graded cohomology ring $H^*(X)$

α^p p -cochain, β^q q -cochain, then

$$(\alpha^p \cup \beta^q)(\sigma) = \alpha^p(\sigma \circ \iota_{0,1,\dots,p}) \cdot \beta^q(\sigma \circ \iota_{p,p+1,\dots,p+q})$$

σ is a singular $(p+q)$ -simplex

$$\text{induces } H^p(X) \times H^q(X) \rightarrow H^{p+q}(X)$$

1. $\cup$ is functorial

If $f: X \rightarrow Y$ is a continuous function,

and $f^*: H^*(Y) \rightarrow H^*(X)$, then

$$f^*(\alpha \cup \beta) = f^*(\alpha) \cup f^*(\beta)$$

2. Other definition

(1) In de Rham cohomology, the cup product of differential forms is induced by the wedge product

(2) For oriented manifolds, the cup product is dual to intersection

Let M^n be an oriented smooth mfd,

A, B are submfd of codim i and j
intersect transversely, then $A \cap B$ is
a submfd of codim $i+j$

Take the Poincaré pairings $[A]^* \in H^i, [B]^* \in H^j$
then there is the equality

$$[A]^* \cup [B]^* = [A \cap B]^* \in H^{i+j}(X, \mathbb{Z})$$

- Cap product

X topological space

$$\cap : H_p(X; \mathbb{R}) \times H^q(X; \mathbb{R}) \rightarrow H_{p-q}(X; \mathbb{R})$$

Singular homology/coho

$$b : p\text{-chain} \quad \psi : q\text{-cochain} \quad q \leq p$$

$$b \cup \psi = \psi(b|_{[v_0, \dots, v_{q-1}]}) b|_{[v_q, \dots, v_p]}$$

if $q > p$
 $\cap$ is zero

- Poincaré duality

M^n a closed oriented mfd

there is a canonical iso $H^k(M, \mathbb{Z}) \rightarrow H_{n-k}(M, \mathbb{Z})$

$$\alpha \mapsto [M] \cap \alpha$$

Rmk: When M is not ~~can~~ compact,

one has to replace homology by Borel-Moore

$$\text{homology } H^i(X) \xrightarrow{\sim} H_{n-i}^{BM}(X)$$

or replace cohomology by cohomology with compact support

$$H_c^i(X) \xrightarrow{\sim} H_{n-i}(X)$$

Rmk: If X is singular $n[X] : H^{n-i}(X) \rightarrow H_i(X)$
 is not necessarily iso

- Poincaré duality (another version)

The operation of intersection define a product

$$H_i(X) \times H_j(X) \xrightarrow{\cap} H_{i+j-n}(X)$$

Poincaré duality, if $i+j=n$, then

$$H_i(X) \times H_j(X) \xrightarrow{\cap} H_0(X) \rightarrow \mathbb{Z}$$

is nondegenerate.

Intersection homology theory

I. the intersection homology group

1. pseudomanifolds and stratification

a compact space X for which

Stratification of X is a filtration by closed subspace $X = X_0 \supset X_{n-1} = X_{n-2} = \Sigma \supset X_{n-3} \supset \dots \supset X_0$

s.t. ---

Rmk: every pseudo admits a stratification

2. perversity is a sequence of integers

$\hat{p} = (p_2, p_3, \dots, p_n)$ s.t. $p_{k+1} = p_k$ or $p_k + 1$

3. The group $C_i(X)$ of all piecewise linear (P.L.) geometric chains is the direct limit under refinement of the $C_*^T(X)$ over all triangulation of X

4. $IC_i^{\hat{p}}(X)$ is the subgroup of $C_i(X)$ consisting of those chains β s.t. $|\beta|$ is $(\hat{p}, i)$ -allowable and $|\partial\beta|$ is $(\hat{p}, i-1)$ -allowable.

5. $IH_i^{\hat{p}}(X)$ is the i th homology group of the chain complex $IC_*^{\hat{p}}(X)$

Rmk: $IH_i^{\hat{p}}(X)$ is f.g and independent of stratification of X

Rmk: $\forall \hat{p}, \quad IH_0^{\hat{p}}(X) = H_0(X - \Sigma) \cong H^n(X)$

$IH_n^{\hat{p}}(X) = H_n(X) \cong H^0(X - \Sigma)$

6. The map to homology and the map from cohomot.
cohomology

Poincaré duality map

$$\cap [X]: H^{n-i}(X) \rightarrow H_i(X) \text{ (not necessarily iso!)}$$

We have a homomorphism

$$H^{n-i}(X) \xrightarrow{\alpha} IH_i^{\hat{p}}(X) \xrightarrow{\omega} H_i(X)$$

$\omega_{\hat{p}}: IH_i^{\hat{p}}(X) \rightarrow H_i(X)$ is induced from the inclusion $IC_{\hat{p}}^{\hat{p}}(X) \subset C_*(X)$

$\alpha_{\hat{p}}$ construction is a bit complicated

We can check $\omega_{\hat{p}} \circ \alpha_{\hat{p}}$ is the Poincaré duality map.

II. the intersection product.

We can construct pairings

$$IH_i^{\hat{p}}(X) \times IH_j^{\hat{q}}(X) \rightarrow IH_{i+j-n}^{\hat{r}}(X)$$

whenever $\hat{p} + \hat{q} \leq \hat{r}$

$[C \cap D] = [C] \cap [D]$ for every dimensionally transverse pair of cycles

III. Thm (Generalized Poincaré duality)

Let $\bar{e}: IH_0^{\bar{e}}(X) \rightarrow \mathbb{Z}$ which counts points with multiplicities ($\bar{e} = (0, 1, \dots, n-2)$)

If $i+j=n$ and $\bar{p} + \bar{q} = \bar{e}$, then the augmented intersection pairing

$$IH_i^{\bar{p}} \times IH_j^{\bar{q}} \rightarrow IH_0^{\bar{e}}(X) \rightarrow \mathbb{Z}$$

is nondegenerate if these groups are tensored with $\mathbb{Q}$

IV. normal space

Def (The normal pseudomanifold X)

(Using Zariski's Main Thm, a normal complex algebraic variety is a normal pseudomanifold)

Thm: $\pi: \bar{X} \rightarrow X$ normalization, $\pi_*: \mathcal{O}_X(\bar{X}) \rightarrow \mathcal{O}_X(X)$ induces iso $\mathrm{IH}_*^{\bar{P}}(\bar{X}) \cong \mathrm{IH}_*^{\bar{P}}(X) \quad \forall \bar{P}$

Thm: ① If X is normal, $\alpha_{\bar{0}}: H^{n-i}(X) \rightarrow \mathrm{IH}_i^{\bar{0}}(X)$ is an iso

② If X is normal, then $\omega_{\bar{1}}: \mathrm{IH}_i^{\bar{1}}(X) \rightarrow H_i(X)$ is an iso

Thm: The intersection pairing

$$\mathrm{IH}_*^{\bar{0}}(X) \times \mathrm{IH}_*^{\bar{0}}(X) \rightarrow \mathrm{IH}_*(X)$$

$$\mathrm{IH}_*^{\bar{0}}(X) \times \mathrm{IH}_*^{\bar{1}}(X) \rightarrow \mathrm{IH}_*(X)$$

coincide with the ~~cc~~ cup and cap product of $\bar{X}$.

IV. the signature

Def: (signature)

If $\dim(A) = 4k$, let $\bar{m} = (0, 0, 1, 1, \dots, 2k-1)$

$\bar{n} = (0, 1, 1, 2, 2, \dots, 2k-1)$, then $\bar{m} + \bar{n} = \bar{e}$.

$\bar{e} = (0, 1, 2, \dots, 4k-2)$

Thus $\mathrm{IH}_{2k}^{\bar{m}}(A) = \mathrm{IH}_{2k}^{\bar{n}}(A)$ and by generalized Poincaré duality, the pairing

$$\mathrm{IH}_{2k}^{\bar{m}}(A) \times \mathrm{IH}_{2k}^{\bar{n}}(A) = \mathrm{IH}_{2k}^{\bar{m}} \times \mathrm{IH}_{2k}^{\bar{n}}(A) \rightarrow \mathbb{Z}$$

is symmetric and nonsingular over $\mathbb{Q}$.

The index of this quadratic form is defined to be the signature of A .

e.g. a complex projective variety X of complex dim $2r$
 $(\dim_{\mathbb{R}}(X) = 4r)$,

$IH_{2r}^{\bar{m}}$ is the middle-perversity intersection homology

V1 sheaf-theoretic approach to intersection homology.

Recall: a bilinear non-degenerate pairing

$$IH_i^{\bar{p}}(X, \mathbb{Q}) \times \underbrace{I^{BM}H_{n-i}^{\bar{q}}(X, \mathbb{Q})}_{(\text{a bit different?})} \rightarrow \mathbb{Q}$$

$$\bar{p} + \bar{q} = \bar{r} = (0, \dots, n-2)$$

Aim: construct a complex of A -sheaves $IC_{\bar{p}}^{\bullet}$
 whose $H^{-i}(X; IC_{\bar{p}}^{\bullet})$ calculates $I^{BM}H_i^{\bar{p}}(X; A)$
 and $H_c^{-i}(X; IC_{\bar{p}}^{\bullet})$ calculates $IH_i^{\bar{p}}(X; A)$.

Rmk: we can also construct Deligne complex
 s.t. $IC_{\bar{p}}^{\bullet}$ is quasi-iso to it and Deligne
 complex doesn't depend on the underlying
 PL structure of X

Rmk: When X is normal, $IC_0^{\bullet} \cong \underline{A} \times [n]$

$$\begin{aligned} \Rightarrow I^{BM}H_i^{\bar{0}}(X; A) &\cong H^{-i}(X, IC_0^{\bullet}) \\ &\cong H^{-i}(X, \underline{A} \times [n]) \cong H^{n-i}(X; A) \end{aligned}$$

Thm: $\mathcal{D}IC_{\bar{q}}^{\bullet}[n] \cong IC_{\bar{p}}^{\bullet}$, if $\bar{q} + \bar{p} = \mathbb{E}$

Cor: $I^{BM}H_i^{\bar{p}}(X; A) \otimes I^{BM}H_{n-i}^{\bar{q}}(X; A) \rightarrow A$
is nondegenerate

VII Rational homology mfd ($\mathbb{Q}$ -mfd)

Def: A topo space X is called a $\mathbb{Q}$ -mfd of real dim m if $\forall x \in X$,

$$H_i(X, X-x; \mathbb{Q}) = \begin{cases} \mathbb{Q} & i=m \\ 0 & i \neq m \end{cases}$$

e.g. orbifolds (i.e. varieties that are locally finite quotients) are $\mathbb{Q}$ -manifold.

Thm. Let X be a complex alg variety of complex dim n and be a $\mathbb{Q}$ -mfd.

Then the middle-perversity $IC_m^{\bullet}$ with $\mathbb{Q}$ -coeff is quasi-iso to $\underline{\mathbb{Q}}_X[2n]$

$$IC_x^B := IC_m^{\bullet}[-2n] \quad IC_x := IC_m^{\bullet}[-n] \Rightarrow \mathcal{D}(IC_x) \cong IC_x$$

Def: Let X be the variety as above

$$\begin{aligned} IH^k(X; \mathbb{Q}) &:= IH^k(X; IC_x^B) \cong I^{BM}H_{2n-k}^m(X; \mathbb{Q}) \\ &= IH^k(X; IC_x[-n]) \end{aligned}$$

$$\begin{aligned} \text{Then } IH^k(X; \mathbb{Q}) &= IH^k(X; IC_x^B) = IH(X, \underline{\mathbb{Q}}_X) \\ &= H^k(X; \mathbb{Q}) \end{aligned}$$

$$\alpha_x: \underline{\mathbb{Q}}_X[n] \xrightarrow{\text{quasi-iso}} IC_x$$

Using Verdier dualizing functor $\mathcal{D}$ to α_x
and $\mathcal{D}(\underline{\mathbb{Q}}_X[n]) \cong \mathcal{D}_x^{\bullet}[-n]$, $\mathcal{D}(IC_x) \cong IC_x$,

$$\text{one get } \underline{\mathbb{Q}}_X \xrightarrow{\text{quasi-iso}} IC_x[-n] \xrightarrow{\text{quasi-iso}} \mathcal{D}_x^{\bullet}[-2n]$$

Applying hypercohomology

$$H^k(X) \xrightarrow{\sim} IH^k(X) \xrightarrow{\sim} H_{2n-k}^{BM}(X)$$

Rmk: for a complex algebraic variety X , we always only consider the middle-perversity $\bar{m}$

$\downarrow$
self-dual

Def: $f: X \rightarrow \text{pt}$, $\mathcal{D}_X^\bullet := f^! \underline{A}_{\text{pt}}$ constant sheaf A

The Verdier dual of a bounded complex

$\mathcal{F}^\bullet \in D^b(X)$ is defined as

$$\mathcal{D}_X \mathcal{F}^\bullet := R\mathcal{H}om^\bullet(\mathcal{F}^\bullet, \mathcal{D}_X^\bullet)$$

V122 hard Lefschetz Thm for IH

Thm: Let X be a complex proj var of pure dim n , $L \in H^2(X, \mathbb{Q})$ the first chern class of an ample line bundle. Then

$$L^i: IH^{n-i}(X; \mathbb{Q}) \xrightarrow{\sim} IH^{n+i}(X; \mathbb{Q})$$

$\forall i \geq 0$, induced by the cup product of L^i

Rmk: Taking $f: X \rightarrow \text{pt}$ in the relative hard Lefschetz is our Thm.

$$L \in H^2(X, \mathbb{Q}) \quad L: \underline{\mathbb{Q}}_X \rightarrow \underline{\mathbb{Q}}_X[2]$$

$$\otimes IC_X \xrightarrow{\sim} L: IC_X \rightarrow IC_X[2]$$